

National Tsing Hua University
1130IEEM 513600
Deep Learning and Industrial Applications
Homework 3

Name: 王昀萱

Student ID: 113030502

1. Download the MVTec Anomaly Detection Dataset from Kaggle. Select one type of product from the dataset. Document the following details about your dataset:

Number of defect classes	4 (不含 good)
Types of defect classes	Bent,color,flip,scratch
Number of images used in your dataset	21-24 per class
Distribution of training and test data	219/110
Image dimensions	700*700

2. Implement 4 different attempts to improve the model's performance trained on the dataset you choose in previous question. Ensure that at least one approach involves modifying the pre-trained model from TorchVision. Summarize the outcomes of each attempt, highlighting the best performing model and the key factors contributing to its success. You may also need to describe other hyperparameters you use in your experiment, like epochs, learning rate, and optimizer.

在這邊我生成了四個模型，用這四種方式分別增加與改善：

1. 模型 A：

用 WeightedRandomSampler 進行 oversampling 來處理類別資料不平衡問題。

(batch size = 32, lr = 0.001, epoch = 50)

Test accuracy is 52.2%

2. 模型 B：

使用資料增強（翻轉、亮度變化、旋轉）來提升泛化能力。

(batch size = 32, lr = 0.001, epoch = 50)

Test accuracy is 56.5%

3. 模型 C：

將 ResNet18 的結構從原本的 fc 換成多層的 MLP 並加入 Dropout，去抑制 overfitting 的問題。

(batch size = 32, lr = 0.001, epoch = 50)

Test accuracy is 65.2%

4. 模型 D：

調整超參數 (batch size = 16)。

(batch size = 16, lr = 0.001, epoch = 50)

Test accuracy is 73.9%

模型 C 是提升最高的模型，這代表修改預訓練的模型架構是最關鍵的改善方式，而影響效能的主因包含了資料平衡處理、資料多樣性與強化分類器的表現。

3. In real-world datasets, we often encounter long-tail distribution (or data imbalance). In MVTec AD dataset, you may observe that there are more images categorized under the 'Good' class compared to images for each defect class.

(i) Define what is 'long-tail distribution'.

在現實中的資料常會有極端不平衡的現象，常見但少量的類別與罕見卻大量的類別，兩者資料數量會形成非常懸殊的差距，此為長尾分布，為二八法則的一種。

(ii) Identify and summarize a paper published after 2020 that proposes a solution to data imbalance. Explain how their method could be applied to our case.

論文名稱：**"Equalization Loss v2: A New Gradient Balancing Loss for Long-tailed Object Detection" (CVPR 2021)**

這篇針對物件偵測中的長尾分佈提出 Equalization Loss v2 (EQLv2)，此方法不直接重加權尾部類別的樣本數，而是調整正負梯度的權重，使得模型在訓練時逐漸強化尾部類別的學習效果，避免過度擬合頭部類別。

如果使用在有關 MVTec AD 的情境，可以搭配 ResNet 進行特徵分類時，利用梯度平衡讓模型更有效學習少數的 defect 類別去提升模型對少數資料的辨識能力，改善長尾分佈帶來的偏誤。

4. The MVTec AD dataset's training set primarily consists of 'good' images, lacking examples of defects. Discuss strategies for developing an anomaly detection model under these conditions.

當 MVTec AD 資料只有正常樣本時，可以使用無監督或單類學習方法，或

是使用預訓練模型提取特徵，計算測試圖像與正常圖像之間在特徵空間的距離來偵測異常。有額外相似資料的話也可以進行transfer learning提升效果。

5. For the task of anomaly detection, it may be advantageous to employ more sophisticated computer vision techniques such as object detection or segmentation. This approach will aid in identifying defects within the images more accurately. Furthermore, there are numerous open-source models designed for general applications that can be utilized for this purpose, including YOLO-World ([website](#)) and SAM ([website](#)).

- (i) To leverage these powerful models and fine-tune them using our dataset, it is necessary to prepare specific types of datasets. What kind of data should be prepared for object detection and for segmentation.

物件偵測需要含有缺陷的影像與對應的 bounding boxes而影像分割需要對每張圖片每個像素進行分類，並標示哪些區域屬於 defect。

- (ii) Why are these models suitable for fine-tuning for our customdataset?

YOLO-World 和 SAM 是大模型，具強大特徵提取能力，支援少量資料微調 (fine-tuning)。YOLO-World 支援文字提示偵測，SAM 能精準切出任意物體，皆適合用於自訂缺陷辨識。代表只要符合能定義出缺陷型態以及標示缺陷類別，即可以作為一份自訂的資料集，泛用性極高。另外transfer-learning，因為model已經過預訓練，所以有高辨識形狀和邊緣的能力，這也是適合fine-tuning的原因。